

L16

SEO Basics

How Search Engines Work, and How Meta Tags Talk to Them

Read this document start to finish, in order. Every section builds on the one before it — by the time you reach the tags themselves, you won't be memorizing syntax, you'll be answering a question you already understand the reason for.

KEY POINT

After reading these notes, you should be able to explain what SEO actually is, describe the three steps a search engine goes through (crawl → index → rank), and confidently write a correct <title>, meta description, and meta keywords tag for any page you build — including your own portfolio project.

Part 1 — What Is SEO, Really?

Before any code, start with a picture in your head.

Priya opens a small bakery in a huge city. Her cakes are genuinely the best in the city. But there's no signboard outside her shop, and she isn't listed in any local directory or on Google Maps. Result: nobody can find her, no matter how good the cakes are.

Now imagine two things change:

1. She puts up a clear signboard outside: “Priya’s Bakery — Fresh Cakes & Pastries.” Anyone walking past instantly knows what the shop is.
2. She gets listed correctly in the local business directory, under the right category, with a short one-line description of what she sells.

She hasn't changed her cakes at all. She has simply made it possible for people — and the systems that help people find businesses — to correctly understand and find her shop.

That is exactly what SEO is, for a website.

Definition

SEO (Search Engine Optimization)

The practice of structuring and describing your web page so that a search engine can correctly understand what the page is about, and show it to the right people when they search for related things.

KEY POINT

You are not tricking Google. You are handing it clean, correct information about your page so it can do its job properly. Good SEO never means hiding information from users to please a search engine — it means making your page's purpose obvious to both humans and machines.

A Misconception to Clear Up Early

SEO does not mean “adding as many popular keywords as possible so Google ranks you higher.” That approach is called keyword stuffing, and it actively gets pages penalized. Real SEO is about clarity, not tricks — you'll see exactly why when we reach the meta keywords tag in Part 5.

Part 2 — How Do Search Engines Actually Work?

Before writing a single tag, you need the mental model of what happens on Google's side of a search. It happens in three stages: crawling, indexing, and ranking.

2.1 Crawling

Crawling

The process where a search engine sends out automated programs — called crawlers, bots, or spiders (Google's is called Googlebot) — that visit web pages, read their content, and follow the links on them to discover more pages.

Analogy: a crawler is like a very fast, very thorough delivery person who visits every shop on every street, notes down what each one sells, and then walks into the next shop the first one pointed to.

Example: if your portfolio page links to your GitHub profile, and GitHub links to your other projects, Googlebot can follow that trail from page to page.

2.2 Indexing

Indexing

What happens after crawling — the search engine stores what it found about your page in a giant database (the index), along with information about the topics, words, and structure the page contains.

Analogy: the delivery person goes back to the office and updates a giant catalog — “Shop X on Street Y sells fresh cakes, open till 9pm, category: bakery.”

Important: a page can be crawled but still not indexed — for example, if it explicitly tells search engines not to index it, or looks low-quality or duplicate. Being indexed is the minimum requirement to ever appear in search results at all.

2.3 Ranking

Ranking

When someone types a search query, the engine looks through everything relevant in its index and orders those pages using hundreds of signals — relevance to the query, page quality, loading speed, mobile-friendliness, links from other sites, and more — to decide the order they appear in.

Analogy: 40 bakeries are listed in the catalog, and someone asks “best chocolate cake near me.” The directory doesn't show all 40 in random order — it ranks them by relevance and quality.

The Pipeline, In One Line

Stage	What Happens	Bakery Analogy
1. Crawl	Bots visit your page and read its HTML.	Delivery person walks into every shop.
2. Index	The page's content and structure get stored in the search engine's database.	The catalog back at the office gets updated.
3. Rank	For a specific search query, indexed pages are ordered by relevance and quality.	Directory shows the best-matching bakeries first.

Where do meta tags fit into this? They don't control ranking directly (with one exception, the title tag). But they are among the clearest signals you give during crawling and indexing about what your page is — and they directly control what a human sees on the results page, which decides whether that person clicks through to you at all.

Part 3 — Why This Matters to You as a Developer

You already know, from earlier lessons, how a browser requests a page and how the DOM is built out of HTML tags. Here is the one sentence that connects all of that to SEO:

KEY POINT

A crawler reads your HTML the same way your browser does before it paints anything on screen — it parses tags and text. It does not take a screenshot of your page and “look” at it the way a person does.

This has real implications for how you build pages:

- If information exists only as text baked into an image (like a banner image with the shop name drawn inside the picture file), a crawler mostly cannot read it. Real text in real tags is what gets read.
- If your page has no `<title>` tag, the crawler has no reliable way to know what to even call your page in search results.
- Structure — headings, semantic tags, meta tags — isn't just for organizing your code. It is literally the language you use to talk to a crawler.

Everything from here on is just one question, answered five different ways: how do I write this so a crawler reads it correctly?

Part 4 — Where These Tags Live: The `<head>` Section

Quick recap: every HTML document has two main parts inside `<html>`:

- **`<head>`** — metadata about the page. Not shown directly on the page itself, but read by the browser and by crawlers.
- **`<body>`** — the actual visible content the visitor sees.

Every tag covered in this document lives inside `<head>`.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- everything in this document goes here -->
</head>
<body>
  <!-- the visible page goes here -->
</body>
</html>
```

Part 5 — The Core Meta Tags

For each tag below: what it does, why it matters for SEO, its syntax, and a real example. Every example uses the same running case — Priya's Bakery — so you can see how the tags work together for one real page.

Meta tags are HTML tags that describe a web page *about* itself, rather than displaying content on the page. They live inside `<head>`, are invisible to visitors, and exist purely to hand information to browsers and search engines — things like what encoding the text uses, how the page should scale on mobile, and what the page is about.

5.1 `<meta charset="UTF-8">`

What it does

Tells the browser and crawler which character encoding the page uses, so text — including special characters, emojis, currency symbols, and non-English scripts — displays correctly instead of turning into garbled symbols.

Why it matters for SEO: wrong or missing encoding can make a crawler misread your text entirely, or make your content render as broken characters for users — which hurts both usability and how confidently a crawler can index the page.

UTF stands for **Unicode Transformation Format** — it's the method used to turn characters into actual stored bytes.

To make sense of that, it helps to separate two things that sound similar but aren't:

- **Unicode** is the master list. It assigns every character in every language — plus emojis and symbols — a unique number, called a **code point**. For example, the letter "A" is Unicode number 65. The 🍰 emoji has its own number too. Unicode itself is just the *assignment* — "A" = 65 — it doesn't say how to actually store that number as bytes on a computer.
- **UTF** is the *encoding* — the actual rule for writing those Unicode numbers down as bytes so a computer can save or send them.

```
<meta charset="UTF-8">
```

Rule: this should be the very first tag inside `<head>`.

5.2 <meta name="viewport" ...>

What it does

Tells the browser how to control the page's dimensions and scaling on different screen sizes — this is what makes a page render properly on a phone instead of showing a tiny, zoomed-out desktop layout.

Why it matters for SEO: Google has used mobile-first indexing for years — it primarily looks at the mobile version of your page to decide rankings, because most searches now happen on phones. A page without this tag looks broken on mobile, and that hurts your ranking directly.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

name="viewport" — the **viewport** is the visible area of a web page inside the browser window (or, on mobile, the actual screen). This attribute tells the browser: "the settings coming next control how the viewport behaves," as opposed to **name="description"** or **charset**, which control different things.

content="width=device-width, initial-scale=1.0" — this is where the actual instructions live, written as comma-separated **key=value** pairs.

- **width=device-width** — sets the viewport's width to match the device's actual screen width, in CSS pixels. Without this, mobile browsers historically assumed every page was built for a desktop screen (defaulting to a fixed width, often 980px) and just shrank the whole thing down to fit — which is why old, un-optimized sites look tiny and forced you to pinch-zoom to read anything. This one value tells the browser: "don't assume desktop — just use the real screen width."
- **initial-scale=1.0** — sets the zoom level the page starts at. 1.0 means a 1:1 ratio — one CSS pixel equals one device pixel, i.e., no zoom applied. Without it, even with the correct width set, some browsers still apply a default zoom that makes things look off on first load.

5.3 <title> — the single most important SEO tag

What it does

Sets the page's title. This is: (a) the big blue clickable headline in Google search results, (b) the text on the browser tab, and (c) what shows when someone bookmarks or shares the link.

Why it matters: the title is the one tag on this list that search engines have confirmed factors directly into ranking — it's treated as the strongest single signal for what a page is about. It's also the first thing a human sees in results, so it has to be accurate but also good enough to make someone want to click.

Length rule: roughly 50–60 characters. Google displays only about the first ~580 pixels of a title (which works out to roughly 50–60 characters depending on the letters used) — anything beyond that gets cut off with "...".

```
<title>Priya's Bakery — Fresh Cakes & Pastries in Pune</title>
```

Good vs. bad, at a glance:

- **Good:** specific and human-readable — "Priya's Bakery — Fresh Cakes & Pastries in Pune." Describes the actual page.

- **Bad (too vague):** “Home” or “Welcome to my website” — tells a crawler and a human nothing.
- **Bad (keyword-stuffed):** “Bakery Cake Pastry Bread Sweets Best Bakery Pune Cake Shop” — reads as spam, and modern Google actively distrusts pages written like this.

Here's how the title and description tags actually show up on a real results page:

• www.priyasbakery.com

Priya's Bakery — Fresh Cakes & Pastries in Pune

Handmade cakes, pastries, and custom orders in Pune. Order online or visit us — fresh baked daily, since 2019.

Figure 1 — A well-written title and description. Both are within length limits and read like a real sentence, not a keyword list.

• www.priyasbakery.com

Priya's Bakery Pune Best Cake Shop Fresh Bread Pastry Custom Cakes Order On...

No information is available for this page.

Figure 2 — What happens without care: the title is keyword-stuffed and gets cut off mid-sentence, and with no meta description, Google has to guess — often showing a placeholder or a random, messy sentence pulled from the page.

5.4 Meta Description

What it does

A short summary of the page's content, shown as the gray text snippet below the title in search results.

Important : the meta description does NOT directly affect your ranking position — Google has confirmed this. But it directly affects your click-through rate (CTR): out of everyone who sees your result, how many actually click it. A page ranked #3 with a compelling description can get clicked more than a page ranked #1 with no description at all. And if you don't write one, Google auto-generates one by grabbing a random chunk of text from your page — which is usually messy and doesn't sell your page well.

Length rule: roughly 150–160 characters. Beyond that, it gets truncated with “...”.

```
<meta name="description" content="Handmade cakes, pastries, and custom orders in Pune. Order online or visit us — fresh baked daily, since 2019.">
```

Writing tips:

- Summarize the page like you're describing it to a friend in one sentence.
- Include what makes this specific page or business distinct — location, specialty, unique offer.
- Give every page on your site a different description — don't copy-paste the same one everywhere.
- Don't stuff keywords here either — write for the human deciding whether to click.

5.5 Meta Keywords — the “historical” tag

What it did

The meta keywords tag was meant to let a page's author explicitly declare which topics or keywords the page is about.

Why it's mostly obsolete: in the early 2000s, site owners abused this tag heavily — stuffing it with hundreds of irrelevant, high-traffic keywords to catch unrelated search traffic (a bakery site listing “movies, free download, celebrity news” just to game rankings). Because the abuse was so widespread, Google officially announced in 2009 that it does not use the meta keywords tag for ranking at all. Most modern crawlers ignore it completely today.

```
<meta name="keywords" content="bakery, cakes, pastries, Pune">
```

VERDICT

Optional today. It doesn't hurt to include it, and a few minor or older search tools might still glance at it — but it has essentially zero real impact on modern search engines. Know that it exists and understand why it lost its power: that history is itself a lesson in why search engines had to stop trusting whatever a site simply claims about itself, and start reading real signals instead.

Part 6 — Putting It All Together: A Full Example

Here is a complete, annotated `<head>` section for Priya's Bakery, using every tag from Part 5 in the order they should appear:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"> ← always first
  <meta name="viewport" content="width=device-width, initial-scale=1.0"> ← mobile scaling

  <title>Priya's Bakery — Fresh Cakes & Pastries in Pune</title> ← ~55 chars

  <meta name="description" content="Handmade cakes, pastries, and
    custom orders in Pune. Order online or visit us — fresh baked
    daily, since 2019."> ← ~150 chars

  <meta name="keywords" content="bakery, cakes, pastries, Pune"> ← optional, legacy
</head>
```

Part 7 — Common Mistakes Beginners Make

Mistake	Why It's a Problem
Leaving <code><title>Document</title></code> from the boilerplate.	The single most common one. Tells users and Google nothing about the page — usually just forgotten after copying a starter template.
Same title and description on every page of a site.	Google can't tell your pages apart, so it may show only one of them and treat the rest as duplicate or lower-value.
Cramming keywords unnaturally into the title or description.	Reads as spam to both humans and Google's spam-detection systems — hurts trust and click-through rate.
Forgetting the viewport tag.	The page looks broken or zoomed on mobile, which hurts user experience and mobile-first ranking.

Mistake	Why It's a Problem
Important text exists only inside an image (e.g., business name baked into a logo graphic).	Crawlers can't reliably read text embedded inside images — that information is effectively invisible to search engines.
Description copy-pasted straight from the page's first paragraph.	Often reads awkwardly out of context and fails as a clear, compelling one-line summary.

Part 8 — Quick Reference Cheat Sheet

Tag	Purpose	Ideal Length	Directly Affects Ranking?
meta charset	Correct text encoding	—	Indirectly (bad encoding hurts indexing)
meta viewport	Mobile-responsive scaling	—	Yes — mobile-first indexing
title	Title shown in results & browser tab	~50–60 characters	Yes — strongest on-page signal
meta description	Summary shown under the title	~150–160 characters	No — but drives click-through rate
meta keywords	Historical topic list	—	No — ignored by Google since 2009

Part 9 — Hands-On Practice

Task 1: Your Own Page

Open your existing portfolio page (from L3 or L14). In its `<head>`, add or fix:

1. `meta charset="UTF-8"` as the very first line inside `<head>`.
2. A proper viewport tag.
3. A specific, descriptive `<title>` — not “Document” or “My Website” — under 60 characters.
4. A meta description of 1–2 sentences, under 160 characters, that would make a stranger want to click.
5. (Optional) a meta keywords tag with 4–5 relevant words.

Self-Check

- ☐ My title tells a stranger exactly what this page is, in under 60 characters.
- ☐ My description reads like a natural sentence, not a keyword list.
- ☐ Every tag is inside `<head>`, before `</head>`.
- ☐ `charset` is the very first tag inside `<head>`.

Task 2 (Bonus): Real-World Check

Open DevTools (right-click → Inspect) on 3 real websites you use often. Look inside their `<head>`. Write down each site's `<title>` and meta description. Notice: do the descriptions read like natural sentences? Are the titles under ~60 characters? This is proof that the exact tags you just learned are running the entire web — not just classroom examples.

Part 10 — Key Takeaways

- SEO is not a trick — it's handing search engines clean, correct information about your page.
- Search engines work in three stages: crawl → index → rank.
- A crawler reads your HTML/markup, not a rendered picture of your page — that's why structure and tags matter.
- `<title>` is the single most important on-page SEO tag — keep it specific and under ~60 characters.
- Meta description doesn't affect ranking directly, but it drives click-through rate — write it for humans.
- Meta keywords is mostly obsolete (ignored by Google since 2009) — know why, but don't obsess over it.

WHAT'S NEXT

These same principles — clear titles, honest descriptions, real semantic structure — come back in Module 6, when you build and ship your final portfolio project. By then, this won't be new information you're memorizing — it'll be a checklist you already understand the reasoning behind.